1. A particle moves such that its velocity is proportional to displacement. The motion is

A) Uniform

B) SHM

C) Circular

D) Random

Answer: B

---

2. If acceleration is zero, velocity is

A) Zero

B) Constant

C) Increasing

D) Decreasing

Answer: B

---

3. Work done by friction is always

A) Positive

B) Negative

C) Zero

D) Infinite

Answer: B

---

4. A body in equilibrium has

A) Net force zero

B) Net torque zero

C) Both

D) None

Answer: C

---

5. If force is doubled, acceleration becomes

A) Same

B) Double

C) Half

D) Zero

Answer: B

---

6. Momentum depends on

A) Mass

B) Velocity

C) Both

D) None

Answer: C

---

7. Kinetic energy is proportional to

A) v

B) $v^2$

C) $v^3$

D) 1/v

Answer: B

---

8. Potential energy at ground level is

A) Maximum

B) Zero

C) Infinite

D) Negative

Answer: B

---

9. Escape velocity is independent of

A) Mass of body

B) Radius

C) g

D) None

Answer: A

10. Orbital velocity is

A) √(GM/R)

B) √(2GM/R)

C) GM/R

D) 2GM/R

Answer: A

---

11. Time period of simple pendulum depends on

A) Mass

B) Length

C) Amplitude

D) Density

Answer: B

---

12. In SHM, acceleration is proportional to

A) x

B) -x

C) v

D) t

Answer: B

---

13. Maximum velocity in SHM occurs at

A) Mean position

B) Extreme

C) Anywhere

D) Zero

Answer: A

---

14. Maximum acceleration in SHM occurs at

A) Mean

B) Extreme

C) Zero

D) Midpoint

Answer: B

---

15. Phase difference between velocity and displacement in SHM is

A) 0

B) $\pi/2$

C) $\pi$

D) $2\pi$

Answer: B

---

16. Sound cannot travel in

A) Air

B) Water

C) Vacuum

D) Solid

Answer: C

---

17. Speed of sound increases with

A) Temperature

B) Pressure

C) Density

D) None

Answer: A

---

18. Doppler effect depends on

A) Relative motion

B) Distance

C) Amplitude

D) Phase

Answer: A

---

19. Electric field lines never

A) Intersect

B) Bend

C) Curve

D) Extend

Answer: A

---

20. Electric potential is scalar because

A) Directionless

B) Has magnitude

C) Has force

D) Has charge

Answer: A

---

21. Capacitance increases with

A) Distance

B) Area

C) Voltage

D) Charge

Answer: B

---

22. Resistance depends on

A) Length

B) Area

C) Material

D) All

Answer: D

---

23. Power in resistor is

A) $I^2R$

B) $V^2/R$

C) VI

D) All

Answer: D

---

24. Magnetic field due to straight wire varies as

A) 1/r

B) r

C) $r^2$

D) Constant

Answer: A

---

25. Force on charge in magnetic field is

A) $qvB \sin\theta$

B) qE

C) qv

D) qB

Answer: A

---

26. Magnetic field inside solenoid is

A) Uniform

B) Zero

C) Infinite

D) Variable

Answer: A

---

27. Induced emf is proportional to

A) Flux

B) Change in flux

C) Current

D) Resistance

Answer: B

---

28. Lenz law indicates

A) Direction

B) Magnitude

C) Energy

D) Voltage

Answer: A

---

29. RMS value of AC is

A) Peak

B) Peak/√2

C) √2 Peak

D) Zero

Answer: B

---

30. Power factor is maximum when

A) $\phi=0$

B) $\phi=90°$

C) $\phi=180°$

D) None

Answer: A

---

31. Total internal reflection occurs when light goes from

A) Dense to rare

B) Rare to dense

C) Same medium

D) Vacuum

Answer: A

---

32. Critical angle depends on

A) Medium

B) Wavelength

C) Intensity

D) Time

Answer: A

---

33. Focal length of concave mirror is

A) Positive

B) Negative

C) Zero

D) Infinite

Answer: B

---

34. Magnification of mirror is

A) -v/u

B) v/u

C) u/v

D) None

Answer: A

---

35. Photoelectric effect depends on

A) Frequency

B) Intensity

C) Distance

D) Time

Answer: A

---

36. Stopping potential depends on

A) Frequency

B) Intensity

C) Area

D) Distance

Answer: A

---

37. De Broglie wavelength inversely proportional to

A) Mass

B) Momentum

C) Velocity

D) All

Answer: D

---

38. Uncertainty principle involves

A) Position & momentum

B) Energy

C) Time

D) Charge

Answer: A

---

39. Energy levels in atom are

A) Continuous

B) Discrete

C) Infinite

D) Random

Answer: B

---

40. Binding energy per nucleon indicates

A) Stability

B) Mass

C) Volume

D) Density

Answer: A

41. Radioactive decay is

A) Spontaneous

B) Forced

C) Controlled

D) Linear

Answer: A

---

42. Half-life is independent of

A) Temperature

B) Pressure

C) Amount

D) All

Answer: D

---

43. Semiconductor conductivity increases with

A) Temperature

B) Pressure

C) Volume

D) Density

Answer: A

---

44. Intrinsic semiconductor has

A) Equal electrons & holes

B) More electrons

C) More holes

D) None

Answer: A

---

45. Diode conducts in

A) Forward bias

B) Reverse bias

C) Both

D) None

Answer: A

---

46. Zener diode is used for

A) Amplification

B) Regulation

C) Switching

D) Oscillation

Answer: B

---

47. Logic gate NOT gives

A) Same

B) Opposite

C) Zero

D) One

Answer: B

---

48. AND gate output is 1 when

A) Any input 1

B) Both 1

C) Both 0

D) None

Answer: B

---

49. OR gate output is 1 when

A) Any 1

B) Both 0

C) Both 1

D) None

Answer: A

---

50. Efficiency of machine is always

A) <100%

B) =100%

C) >100%

D) Infinite

Answer: A

---

51. Friction depends on

A) Normal reaction

B) Area

C) Velocity

D) None

Answer: A

---

52. Work-energy theorem relates

A) Work & KE

B) Work & PE

C) Force & energy

D) None

Answer: A

---

53. Angular velocity unit is

A) rad/s

B) m/s

C) m

D) s

Answer: A

---

54. Torque unit is

A) Nm

B) N

C) J

D) W

Answer: A

---

55. Angular momentum conserved when

A) No external torque

B) Force zero

C) Energy zero

D) None

Answer: A

---

56. Rolling motion involves

A) Translation

B) Rotation

C) Both

D) None

Answer: C

---

57. Heat is form of

A) Energy

B) Mass

C) Force

D) Work

Answer: A

---

58. Specific heat depends on

A) Substance

B) Mass

C) Volume

D) Pressure

Answer: A

---

59. Carnot efficiency increases with

A) Temperature difference

B) Pressure

C) Volume

D) Work

Answer: A

---

60. Entropy measures

A) Disorder

B) Energy

C) Heat

D) Work

Answer: A

---